

ECON485 Fall 2025
Midterm Exam 21.11.2025 – KEY 1 (PROFESSIONAL)

Student Number		Grading	
Name and Surname		Questions 1	 out of 15
Student Signature	THIS KEY IS PREPARED TO DEMONSTRATE HOW A SEASONED ANALYST WORKS. THERE IS ANOTHER KEY WHICH DEMONSTRATES AI USE.	Question 2	 out of 25
		Question 3	 out of 15
Notes: 1. Exam duration is 150 minutes, please take your time. Some questions and parts are easy to answer once you do the previous ones. In particular, you must repeatedly reuse earlier designs (ERD, normalized tables) when answering Q3 and Q4. 2. All questions are based on the Case Study which includes non-trivial business rules. Please read it carefully. Take notes, draw diagrams or small tables. Use the back of pages if necessary for your notes. These notes will not be graded, so you can make mistakes there. You are expected to make mistakes. 3. Some questions test your skills as an analyst. As analysis is an iterative process, the questions are designed to make you add more and more to your DB design. The remaining questions test your skills in the technical aspects (ie. SQL statements, and discussion of concepts). 4. You are allowed to use your notes and check up on the course Github for previous examples. 5. You are allowed and encouraged to use AI tools we covered in the course. 6. Please write your name on the corner of each page. Also if you need more space, feel free to use the back of pages.		Question 4	 out of 35
		Question 5	 out of 10
		Question 6 (Bonus)	 out of 10
		Question 7 (Bonus)	 out of 10
		Total	 out of .100

CASE STUDY

The Catholic Church served as the primary custodian of vital records throughout medieval Europe, especially from the early Middle Ages until the late fifteenth century, because secular governments were often too weak, fragmented, or illiterate to maintain systematic documentation.

Literacy resided almost entirely with the clergy, and parish priests together with monastic scribes therefore recorded baptisms, marriages, and deaths. In the earlier centuries these entries appeared as informal notes written in the margins of missals, on loose sheets, or inside monastic cartularies that mainly copied charters and privileges.

From about the thirteenth century onwards, many parishes began keeping more organised ledgers, although practices varied widely by region. Only after the Council of Trent in 1563 did the Church mandate standardised, separate registers for baptisms (including godparents' names), marriages, and burials, with duplicate copies sent to the bishop. Most entries were in Latin and remained extremely

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brief – a typical baptismal record might read no more than “John, son of William the smith, baptised on the feast of St Michael, 1350”.

Communication between local churches and Rome followed a strict hierarchy. Everyday matters – ordinations, institutions to benefices, or simple marriage dispensations – were handled by the diocesan bishop and recorded in his register. Only cases involving high-ranking nobles, politically sensitive unions, or impediments in the third or fourth degree of kinship were referred upward, sometimes as far as the papal curia.

Requests travelled by trusted messenger or professional courier, and papal replies arrived as bulls (formal documents with a lead seal), briefs, or ordinary letters. The journey from England or Germany to Rome (or to Avignon during the fourteenth-century papal residence there) could easily take six months to a year, so wealthy families often kept permanent agents at the curia to speed the process.

Marriage required careful scrutiny because the Church prohibited unions within certain degrees of consanguinity (blood relationship) and affinity (relationship by marriage). Before the Fourth Lateran Council of 1215 the prohibition extended to the seventh degree, making almost every noble marriage technically incestuous and in need of dispensation. Innocent III reduced the forbidden degrees to the fourth (first cousins), a rule that has remained in force ever since. Direct-line relationships and sibling marriages were prohibited by divine law and could never be dispensed.

For ordinary parishioners the priest announced banns on three successive Sundays so that anyone aware of an impediment – kinship, pre-contract, or vows of chastity – could come forward.

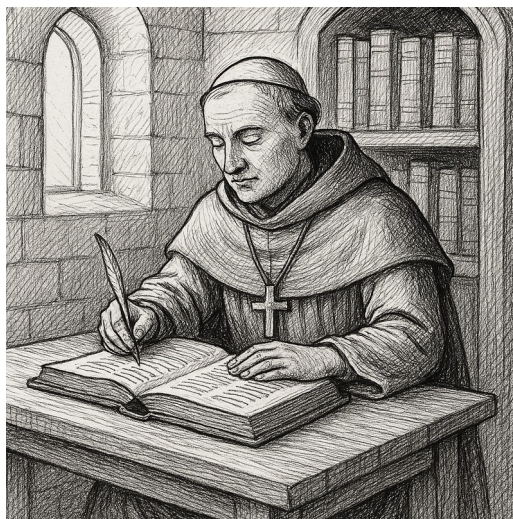


Figure 1: Medieval priest working on marriage records

Noble and royal couples, however, frequently needed dispensations for political alliances, and they submitted detailed petitions (often accompanied by a family tree) to the bishop or directly to Rome, explaining why the union served the peace of Christendom or prevented greater evils. Once granted,

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the dispensation was recorded and the marriage could proceed; without it the union was invalid and any children illegitimate, with devastating consequences for succession.

Beyond baptisms, marriages, and deaths, the medieval Church maintained many other records that affected secular life. Bishops' registers documented the appointment of priests, visitations to parishes, and the proving of wills, and the granting of marriage licences.

Monasteries kept cartularies copying every land grant and privilege, together with necrologies listing the death dates of benefactors so that prayers could be said in perpetuity. Ecclesiastical courts heard cases of defamation, breach of marriage contract, and bastardy, generating voluminous acts that often supply the only evidence of family relationships when parish registers are lost.

Until 1858 in England (and comparable dates elsewhere) the Church retained jurisdiction over probate, so wills and detailed inventories of possessions passed through church courts. Priests frequently acted as clerks for manorial courts, recording land transfers, fines, and inheritance among peasants. Some parishes, especially after Trent, compiled status animarum or "soul books" – household-by-household censuses noting every inhabitant's age and sacramental status – that function as early local population surveys. Tithe lists and taxation records drawn up for church levies provide yet more quasi-censuses.

Taken together, these ecclesiastical documents formed the nearest thing medieval Europe possessed to a centralised population database. A person's baptismal entry proved legitimacy and noble status; the parents' marriage record (or the absence of a valid one) determined whether heirs could inherit land, titles, or thrones; death dates fixed the moment when property changed hands or a new reign began.

Kings, dukes, and barons relied on church archives to defend their claims against rivals, and disputed or forged entries could ignite civil wars. The Church's monopoly on literacy and record-keeping therefore placed it at the very heart of medieval political stability: its registers were simultaneously spiritual, legal, and dynastic instruments upon which the fate of families, estates, and entire kingdoms frequently depended.

-- This space is intended for your note taking --

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Question 1 (Analysis, 15 points)

There is an empty table below about the types of records kept by the church and their purposes. Fill out the table for the the most important five types of records kept by the Church. Hint: There are more than 5, relate importance to the following questions by checking up what's ahead.

A PROFESSIONAL APPROACH IS TO EVALUATE IN-SCOPE CONCEPTS (RECORDS) BASED ON THE FOLLOWING QUESTIONS. THE TABLE SHOULD REFLECT ALL IN-SCOPE CONCEPTS, AND MOST RELATED OUT-OF-SCOPE CONCEPTS.

Record Type	What it contains	Purposes & use cases
Baptismal Records	Individual's name, baptism date, parents' names (including father's profession), and godparents' names.	Proved legitimacy and noble status . Served as early local population surveys.
Marriage Registers	Names of the couple, marriage date, and records of dispensations (if required).	Determined whether heirs could inherit land, titles , or thrones . Enforced prohibitions against consanguinity and affinity
Burial/Death Registers	Deceased person's name, death/burial date .	Fixed the moment when property changed hands or a new reign began.
Bishops' Registers	Appointment of priests, parish visitations, and the granting of marriage licenses and proving of wills.	Documented high-ranking matters like ordinations, benefices, and essential marriage dispensations handled at the diocesan level.
Wills & Inventories	Records of the deceased's last testament , listing possessions and sometimes detailed inventories.	Determined who inherited property (land, titles , thrones). The Church retained jurisdiction over probate until 1858 (in England).

-- The next page is landscape for the size of the table to fit --

Question 2 (Analysis and Core Database Concepts, 20 points)

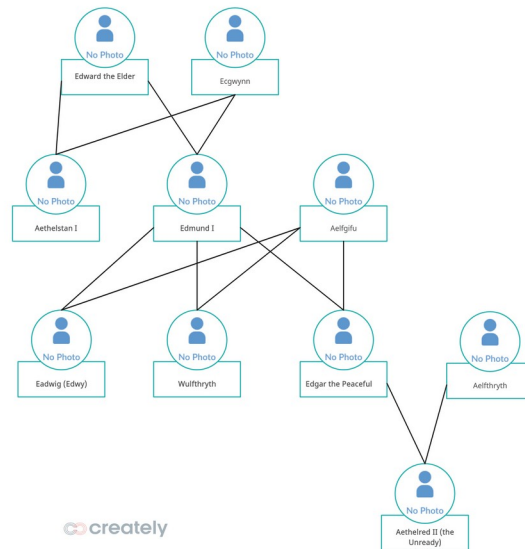
Please study the table below which depicts actual data on Aethelstan I, first real king of England (who united the four kingdoms).

name	mother	father	birth_date	death_date	status	bride	groom	marriage_date	marriage_end	end_reason
Edward the Elder	—	—	871-06-01	924-07-17	noble	—	—	—	—	—
Ecgbrynn	—	—	875-03-15	901-08-10	noble	—	—	—	—	—
Aethelstan I	Ecgbrynn	Edward the Elder	894-05-05	939-10-27	noble	—	—	—	—	—
Eadgifu	—	—	902-01-01	966-08-26	noble	—	—	—	—	—
Edmund I	Eadgifu	Edward the Elder	921-01-01	946-05-26	noble	—	—	—	—	—
Aelfgifu (wife of Edmund I)	—	—	922-07-01	944-01-01	noble	—	—	—	—	—
	—	—	—	—	—	Aelfgifu	Edmund I	939-01-01	944-01-01	death (bride)
Eadwig (Edwy)	Aelfgifu	Edmund I	940-01-01	959-10-01	noble	—	—	—	—	—
Edgar the Peaceful	Aelfgifu	Edmund I	943-01-01	975-07-08	noble	—	—	—	—	—
Wulfthryth	—	—	945-01-01	1000-01-01	noble	—	—	—	—	—
Aelfthryth	—	—	945-01-01	1001-11-17	noble	—	—	—	—	—
	—	—	—	—	—	Aelfthryth	Edgar the Peaceful	964-01-01	975-07-08	death (groom)
Aethelred II (the Unready)	Aelfthryth	Edgar the Peaceful	966-01-01	1016-04-23	noble	—	—	—	—	—

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a) (10 points) Draw the genealogy graph (who married who and who was born out of which marriage) based on this table. Do not bother with writing the names exactly correctly.

WE USE AI TO CREATE A TEXT DATA TABLE (CSV or GEDCOM) AND UPLOAD THAT FILE TO ANY ONLINE GENEALOGY TOOL.



b) (5 points) If we implemented this table in MySQL what would be the data types for these columns,

name, mother, father, status, bride, groom, end_reason - VARCHAR

birth date, death date, marriage_date, marriage_end, - DATE

end_reason – ENUM (death_bride, death_groom) as there may be additional reasons in the future

c) (5 points) write the CREATE statement for this table, with proper data types. You do not need to write the INSERT statements for data population. You are required to demonstrate your command of SQL to complement part (b) above.

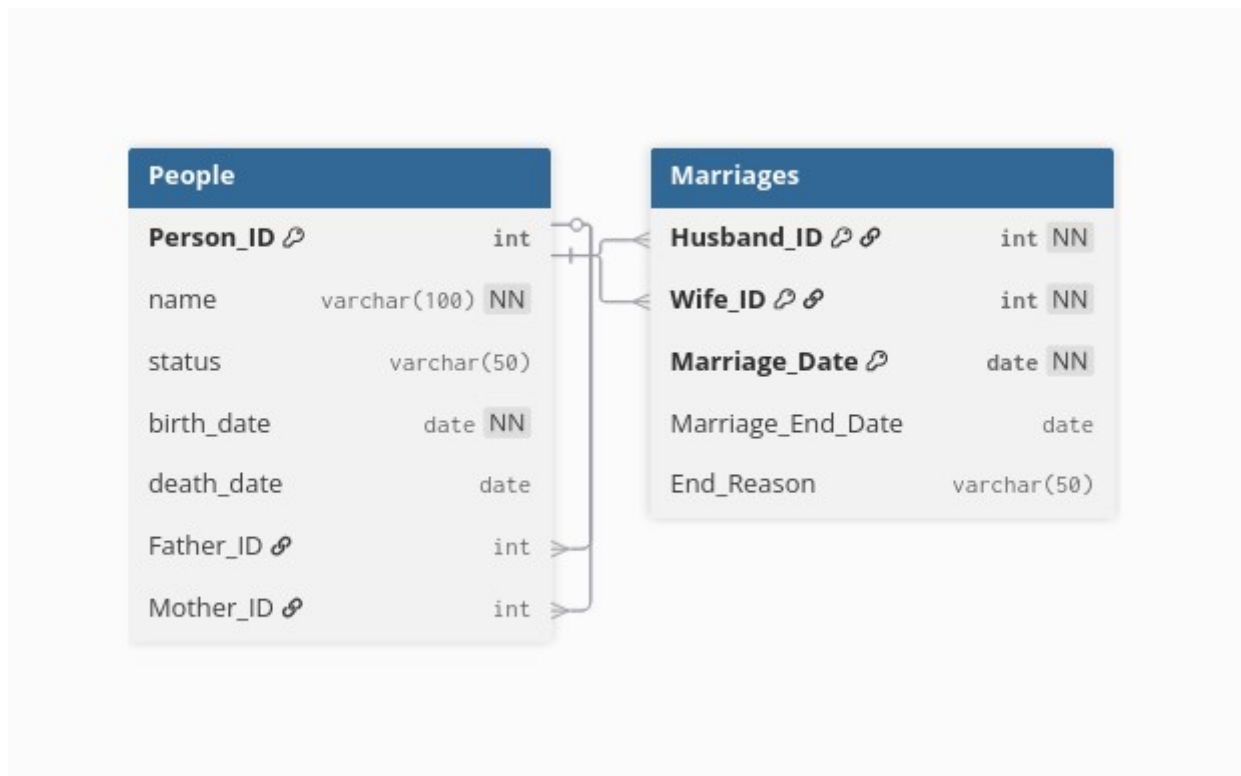
```
CREATE TABLE
Midterm_Lineage_Unnormalized (
-- Personal Data Columns
-- NULL for Marriage Records
name          VARCHAR(100),
mother        VARCHAR(100),
father        VARCHAR(100),
status        VARCHAR(50),
-- Date Columns
-- Using DATE type for efficiency
birth_date    DATE,
death_date    DATE,
```

```
-- Marriage Data Columns
-- NULL for Individual Records
bride         VARCHAR(100),
groom         VARCHAR(100),
marriage_date DATE,
marriage_end  DATE,
-- Categorical field using ENUM
-- for data integrity
end_reason    ENUM('death (bride)', 'death
(groom)')
);
```

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d) (10 points) Please normalize this table to 2NF. Add any primary and foreign keys as you'd like. Draw a pseudo ER diagram on the normalized tables. You are not required to create perfect diagram, just enough to understand the structure.

USING BASIC RULES, NORMALIZATION TO 2NF CREATES TWO TABLES: PEOPLE AND MARRIAGE. USING AI TO CREATE DBML and IMPORTING THAT DBML INTO DBDIAGRAM.IO CREATES THE FOLLOWING ER DIAGRAM.



PLEASE NOTE THAT:

THE PEOPLE TABLE DEMONSTRATES A RECURSIVE ONE-TO-MANY RELATIONSHIP (ALSO KNOWN AS A SELF-REFERENCING RELATIONSHIP) THAT IS FUNDAMENTAL TO MODELING FAMILY TREES.

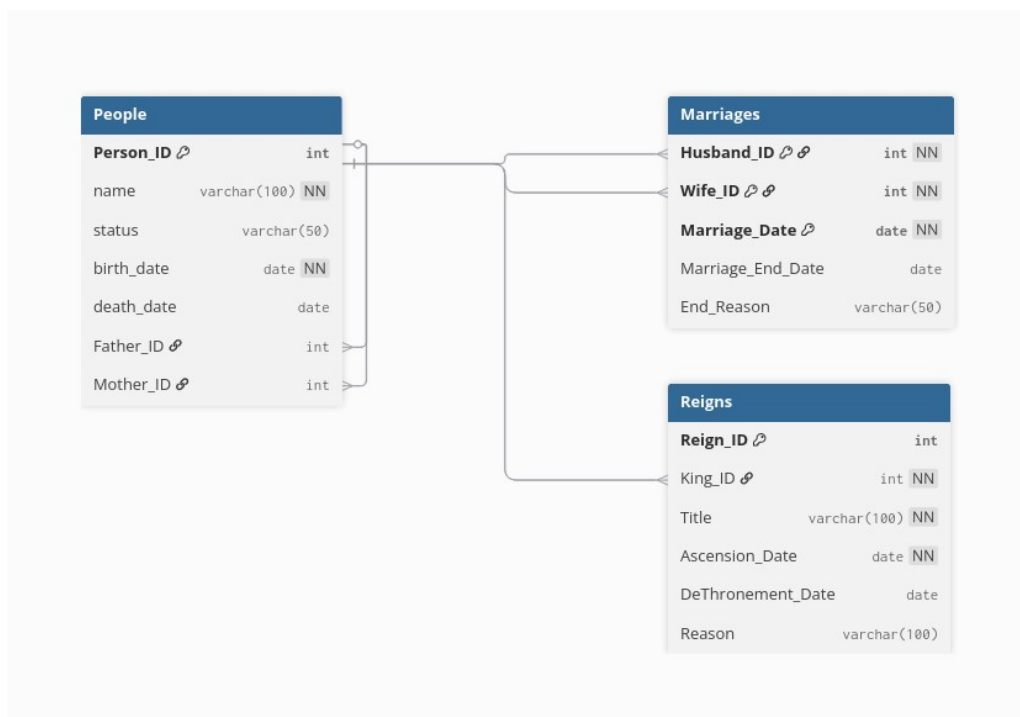
- THE "ONE" SIDE (PARENT/SUPER-TYPE): THE RELATIONSHIP ORIGINATES FROM THE PERSON_ID (PRIMARY KEY) COLUMN. A SINGLE PERSON (THE PARENT) CAN BE REFERENCED MULTIPLE TIMES AS A PARENT.
- THE "MANY" SIDE (CHILD/SUB-TYPE): THE RELATIONSHIP TARGETS THE FATHER_ID AND MOTHER_ID (FOREIGN KEYS) COLUMNS. MANY CHILDREN (ROWS) WILL REFERENCE THE SAME PARENT'S PERSON_ID.
- THEY ELIMINATE DATA REDUNDANCY BY REFERENCING THE PARENT'S UNIQUE PERSON_ID INSTEAD OF STORING THE PARENT'S FULL NAME, BIRTH DATE, AND DEATH DATE REPEATEDLY IN THE CHILD'S RECORD. THIS IS A CORE FEATURE OF A NORMALIZED GENEALOGY MODEL.

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Question 3 (Analysis and Core Database Concepts, 15 points)

Ascension to throne and being de-throned is common and kings' lineage would be very important records as well. The lineage diagram you created illustrates the ascension with two different options. Aethelstan I died without heirs so his brother ascended. But from that point on there were heirs. The diagram does not illustrate the forceful takeover of the throne, but the English history afterwards is very illustrative of how that happens. In rare cases, more than one king was also present (ie. civil war between those multiple kings).

a) (10 points) Extend these tables by creating a kings table. Assume if someone ascends as queen they will count as king as well. Draw the ER diagram for this table, and shows its relations to other tables. You can summarize the other tables in your diagram. You do not need a perfect diagram, just illustrate your table.



b) (5 points) Write a SQL statement (ie. SELECT) that queries the current king(s) based on the table you created.

```
SELECT
    P.name AS Current_Monarch_Name,
    R.Title AS Reign_Title,
    R.Ascension_Date
FROM
    People P
INNER JOIN
    Reigns R ON P.Person_ID = R.King_ID
WHERE
```

```
-- The core condition: a current reign is one
that has no end date defined.
    R.DeThronement_Date IS NULL
-- Optional: Exclude deceased individuals,
although a reigning monarch
-- dying without a successor means the row
won't be NULL for long.
-- AND P.death_date IS NULL
ORDER BY
    R.Ascension_Date;
```

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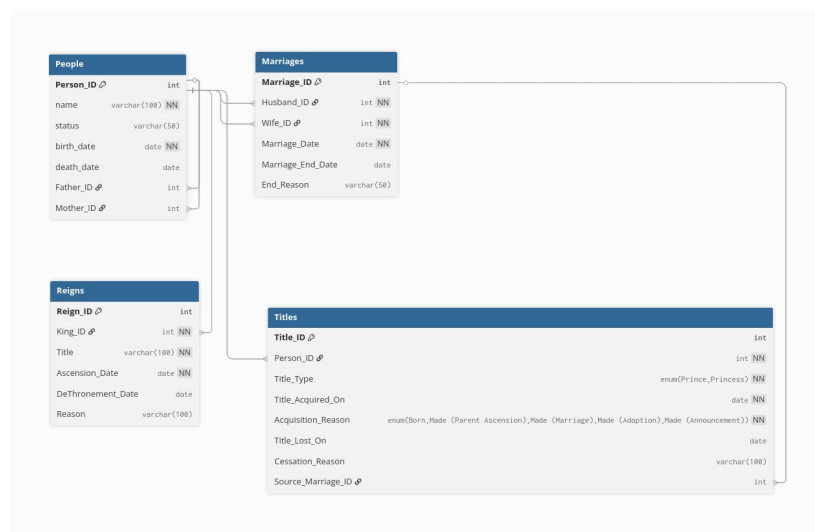
Question 4 (Analysis and Core Database Concepts, 35 points)

Someone could become a prince or princess by birth (as their parent is king at birth) or become one (as their parent becomes a king, or marries into a prince or princess, in rare cases by adoption of king, or also in rare cases by announcement of king).

- If a prince by birth (or adoption) has his parents de-throned, they remain a prince.
- If they become a princess by marriage, and their spouse dies they remain a princess, but they should not re-marry a commoner. If they remarry a commoner their title is removed.
- In contrast princesses who are princess by lineage are always princesses.
- For becoming a prince through marriage, a man who marries a princess who holds the throne in her own right is almost never made prince in the same permanent way a woman is. Their prince status ends with the marriage (divorce, death, etc).
- So there is a distinction between "born" and "made" princes/princesses.

Based on this additional information:

a) (10 points) Extend these tables by creating a princes table to be able to understand who is a prince/princess, since when (at birthdate or at parents' ascension), and why (born or made). Draw the ER diagram for this table, and shows its relations to other tables. You can summarize the other tables in your diagram. You do not need a perfect diagram, just illustrate your table.



b) (5 points) write a SQL statement that creates this table for MySQL.

-- Modified Marriages Table:

```
CREATE TABLE Marriages (
  Marriage_ID INT AUTO_INCREMENT
  PRIMARY KEY,
  Husband_ID INT NOT NULL,
  Wife_ID INT NOT NULL,
  Marriage_Date DATE NOT NULL,
  Marriage_End_Date DATE,
  End_Reason VARCHAR(50),
```

```
-- Ensures no duplicate marriage records
  UNIQUE KEY (Husband_ID, Wife_ID,
  Marriage_Date),
  FOREIGN KEY (Husband_ID)
  REFERENCES People(Person_ID),
  FOREIGN KEY (Wife_ID) REFERENCES
  People(Person_ID)
);
```

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-- New Titles Table

```
CREATE TABLE Titles (
  Title_ID      INT AUTO_INCREMENT
PRIMARY KEY,
  Person_ID     INT NOT NULL,
  -- Title Type (Prince or Princess)
  Title_Type    ENUM('Prince', 'Princess')
NOT NULL,
  Title_Acquired_On  DATE NOT NULL,
  -- Acquisition Reason based on Page 9 of the
exam
```

```
  Acquisition_Reason  ENUM('Born', 'Made
(Parent Ascension)', 'Made (Marriage)', 'Made
(Adoption)', 'Made (Announcement)') NOT
NULL,
  Title_Lost_On      DATE, -- NULL if the title
is currently held
  Cessation_Reason   VARCHAR(100),
  Source_Marriage_ID INT,
  -- Constraints
  FOREIGN KEY (Person_ID) REFERENCES
People(Person_ID),
  FOREIGN KEY (Source_Marriage_ID)
REFERENCES Marriages(Marriage_ID)
);
```

THIS FINAL MODEL USES A SET OF ONE-TO-MANY (1:N) RELATIONSHIPS TO STRUCTURE THE COMPLEX HISTORICAL DATA:

- LINEAGE (PEOPLE TO PEOPLE) [RECURSIVE]: FROM PEOPLE.FATHER_ID TO PEOPLE.PERSON_ID. THE N IS THE CHILD RECORD, AND THE 1 IS THE PARENT RECORD. MANY CHILDREN REFERENCE ONE PARENT. THIS MODELS THE PARENT-CHILD LINK BY ALLOWING A PERSON'S PERSON_ID TO BE REFERENCED AS A FATHER_ID OR MOTHER_ID IN MULTIPLE OTHER ROWS.
- MARRIAGE PARTICIPATION (MARRIAGES TO PEOPLE): FROM MARRIAGES.HUSBAND_ID TO PEOPLE.PERSON_ID AND FROM MARRIAGES.WIFE_ID TO PEOPLE.PERSON_ID.
- SOVEREIGN TITLES (REIGNS TO PEOPLE): FROM REIGNS.KING_ID TO PEOPLE.PERSON_ID. MANY REIGN PERIODS CAN BELONG TO THE SAME ONE INDIVIDUAL. **THIS ACCOMMODATES THE BUSINESS RULE WHERE A MONARCH MIGHT BE DE-THRONED AND THEN REGAIN THE TITLE LATER (NON-CONTIGUOUS REIGNS).**
- NON-SOVEREIGN TITLES (TITLES TO PEOPLE): FROM TITLES.PERSON_ID TO PEOPLE.PERSON_ID. MANY PRINCE/PRINCESS TITLES (E.G., PRINCE BY BIRTH, THEN PRINCE BY SUCCESSION) CAN BELONG TO THE SAME ONE INDIVIDUAL.
- MARRIAGE-CONFERRED TITLES (TITLES TO MARRIAGES): FROM TITLES.SOURCE_MARRIAGE_ID TO MARRIAGES.MARRIAGE_ID. THIS IS A CONDITIONAL FOREIGN KEY. **IF A TITLE IS ACQUIRED BY MARRIAGE (ACQUISITION_REASON = 'MADE (MARRIAGE)'), THE TITLE RECORD MUST LINK TO THE SPECIFIC MARRIAGE_ID THAT CONFERRED THE STATUS.** MANY TITLES (ONE FOR THE MAN IF APPLICABLE, ONE FOR THE WOMAN) CAN BE SOURCED FROM THE SAME ONE MARRIAGE EVENT.

NOTE: THE MARRIAGES TABLE IS NOW A **3NF** STRUCTURE WITH A SURROGATE PRIMARY KEY (MARRIAGE_ID).

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c) (10 points) write a SQL statement that queries the current prince & princesses (ie. alive), their reason (born or married).

```
SELECT
    P.name AS Current_Title_Holder,
    T.Title_Type,
    T.Acquisition_Reason
FROM
    People P
INNER JOIN
    Titles T ON P.Person_ID = T.Person_ID --
Link the person to their titles
WHERE
    -- Condition 1: The person is currently alive
    P.death_date IS NULL
    -- Condition 2: The specific title is currently
    held (has not been lost)
    AND T.Title_Lost_On IS NULL
    -- Condition 3: Filter by the requested
    acquisition reasons (Born or Married)
    AND T.Acquisition_Reason IN ('Born', 'Made
    (Marriage)')
ORDER BY
    T.Title_Type,
    P.name;
```

d) (10 points) write one or more SQL statements that updates the records in case of a "made" princess is re-marrying a commoner and losing princess status.

```
UPDATE
    Titles
SET
    Title_Lost_On = CURRENT_DATE(),
    -- Sets the cessation date to today (or the date of remarriage)
    Cessation_Reason = 'Remarriage to commoner'
WHERE
    Person_ID = TARGET_PRINCESS_ID -- Use the ID of the princess who just remarried
    AND Title_Type = 'Princess'
    AND Acquisition_Reason = 'Made (Marriage)' -- Only target titles acquired via marriage
    AND Title_Lost_On IS NULL; -- Only update currently active titles
```

Question 5 (Core Database Concepts, 10 points)

In Question 4, you used more than one table for your queries. Did your SQL statements make use of joins or they employed simple use of WHERE clause? In any case, explain why you used (or not used) joins.

THE SQL STATEMENT FOR QUESTION 4C DID MAKE USE OF JOINS. JOINS WERE NECESSARY BECAUSE THE DESIGN ADOPTED A NORMALIZED STRUCTURE WHERE BIOGRAPHICAL DATA IS STORED IN THE PEOPLE TABLE, AND TITLE-SPECIFIC DATA IS STORED IN THE TITLES TABLE. WHERE CLAUSE ALONE, USING SIMPLE FILTERING CANNOT ACCESS OR RELATE DATA ACROSS DIFFERENT TABLES. THE INNER JOIN ON THE SHARED KEY, P.PERSON_ID = T.PERSON_ID, ESTABLISHES THE LOGICAL LINK, ALLOWING THE QUERY TO ACCURATELY IDENTIFY A PERSON'S NAME AND SIMULTANEOUSLY CHECK THE STATUS OF THEIR ASSOCIATED TITLE AND ENSURE THEY ARE ALIVE.

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Question 6 (Core Database Concepts, 10 points, Bonus Points)

Assume someone is de-throned (not king anymore) and then by civil war regains the title (is king again). The table you have probably has a single period for being a king.

a) (5 points) Discuss how would you update your kings and princes table(s) to accommodate for this. You do not need to draw ER diagrams, etc. Just discuss verbally. If your table(s) already accommodate this, discuss how it already does.

THE CURRENT REIGNS TABLE ACCOMMODATES THIS SCENARIO BY HAVING A ONE-TO-MANY (1:N) RELATIONSHIP WITH PEOPLE. TO UPDATE FOR A REGAINING OF THE TITLE:

- CLOSE THE FIRST REIGN: UPDATE THE ORIGINAL REIGN RECORD TO SET THE DETHRONEMENT_DATE.
- START THE NEW REIGN: INSERT A NEW ROW INTO THE REIGNS TABLE WITH THE SAME KING_ID, A NEW ASCENSION_DATE, AND A NULL DETHRONEMENT_DATE.

THIS CREATES A CLEAN, NON-OVERLAPPING HISTORICAL RECORD FOR EACH PERIOD OF SOVEREIGNTY.

b) (5 points) Would any previously 1-1 relations become 1-N? And if yes, how would you alter the relation structure? If your table(s) already accommodate this, discuss the existing relations. You do not need to draw ER diagrams, etc. for this part as well.

NO, THE CORE RELATIONSHIP DOES NOT NEED TO BE ALTERED.

THE RELATIONSHIP BETWEEN PEOPLE AND REIGNS WAS ALREADY ESTABLISHED AS A 1:N (ONE PERSON TO MANY REIGNS) IN THE EXTENDED MODEL (Q3A).

THIS EXISTING 1:N STRUCTURE ALREADY ACCOMMODATES THE BUSINESS RULE OF A KING REGAINING A TITLE. NO CHANGES ARE REQUIRED BECAUSE THE FOREIGN KEY REIGNS.KING_ID ALLOWS THE SAME INDIVIDUAL TO BE LINKED TO MULTIPLE, SEPARATE REIGN RECORDS.

Question 7 (Core Database Concepts, 10 points, Bonus points)

Discuss shortly if being in 2NF or 3NF makes adding tables, adding columns or altering 1-1 / 1-N status of relations easier or harder.

3NF MAKES DATA INSERTION HARDER BUT MUCH SAFER BECAUSE IT ELIMINATES TRANSITIVE DEPENDENCIES (WHERE ONE NON-KEY COLUMN DEPENDS ON ANOTHER NON-KEY COLUMN). THIS ENSURES THAT EVERY FACT IS STORED EXACTLY ONCE, PREVENTING INCONSISTENCIES AND UPDATE ANOMALIES. WHILE 2NF IS SIMPLER, IT STILL ALLOWS REDUNDANCY. THE MAJOR ADVANTAGE OF 3NF IS THAT WHEN YOU INSERT A NEW RECORD, YOU ONLY NEED TO PROVIDE THE PRIMARY KEY AND THE ATTRIBUTE THAT DIRECTLY DEPENDS ON IT, MAKING EACH INSERTION ATOMIC AND PROTECTING THE OVERALL INTEGRITY OF THE HISTORICAL DATA STRUCTURE.